PAN-ATLANTIC UNIVERSITY

Computer Networks

Week 4

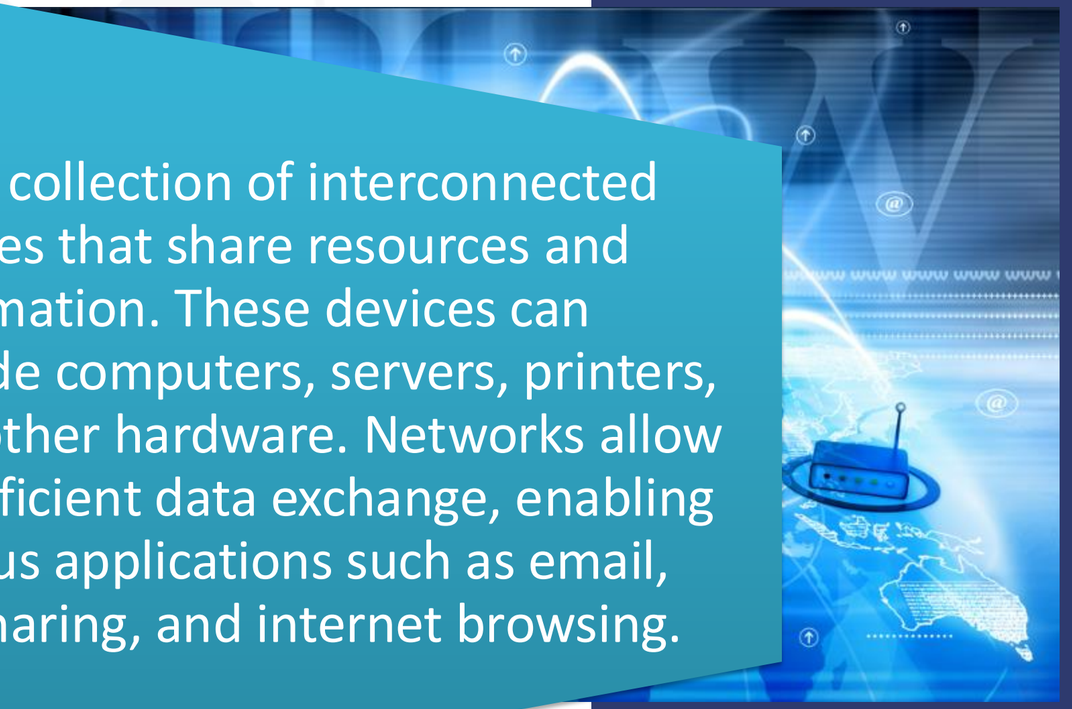
Learning Objectives

- What is a Computer Network?
- Goals of Computer networks
- Types of Computer Networks Architecture
- Computer Network Devices
- Network Protocols
- Internet vs internet
- Intranet vs Extranet
- VPN vs Client/Server Network



Computer Networks

A computer network is a system in which multiple computers are connected to each other to share information and resources. A simple computer network can be built only from two computers while a complex computer network can be built from several thousand computers.



It is a collection of interconnected devices that share resources and information. These devices can include computers, servers, printers, and other hardware. Networks allow for efficient data exchange, enabling various applications such as email, file sharing, and internet browsing.

Goals of Computer Networks

Resource Sharing:	• Many organization have a substantial number of computers in operations, which are located apart. Ex. A group of office workers can share a common printer, fax, modem, scanner, etc. .
High Reliability:	• If there are alternate sources of supply, all files could be replicated on two or more machines. If one of them is not available, due to hardware failure, the other copies could be used.
Inter-Process Communication:	• Network users, located geographically apart, may converse in an interactive session through the network. In order to permit this, the network must provide almost error-free communications.
Flexible Access:	• Files can be accessed from any computer in the network. The project can be begun on one computer and finished on another.
Security	• Computer networks must be secure to protect against unauthorized access, data breaches, and other security threats. This includes implementing measures such as firewalls, antivirus software, and encryption to ensure the confidentiality, integrity, and availability of data.
Performance	• Computer networks must provide high performance and low latency to ensure that applications and services are responsive and available when needed. This requires optimizing network infrastructure, bandwidth utilization, and traffic management.
Scalability	• Computer networks must be designed to scale up or down as needed to accommodate changes in the number of users, devices, and data traffic.

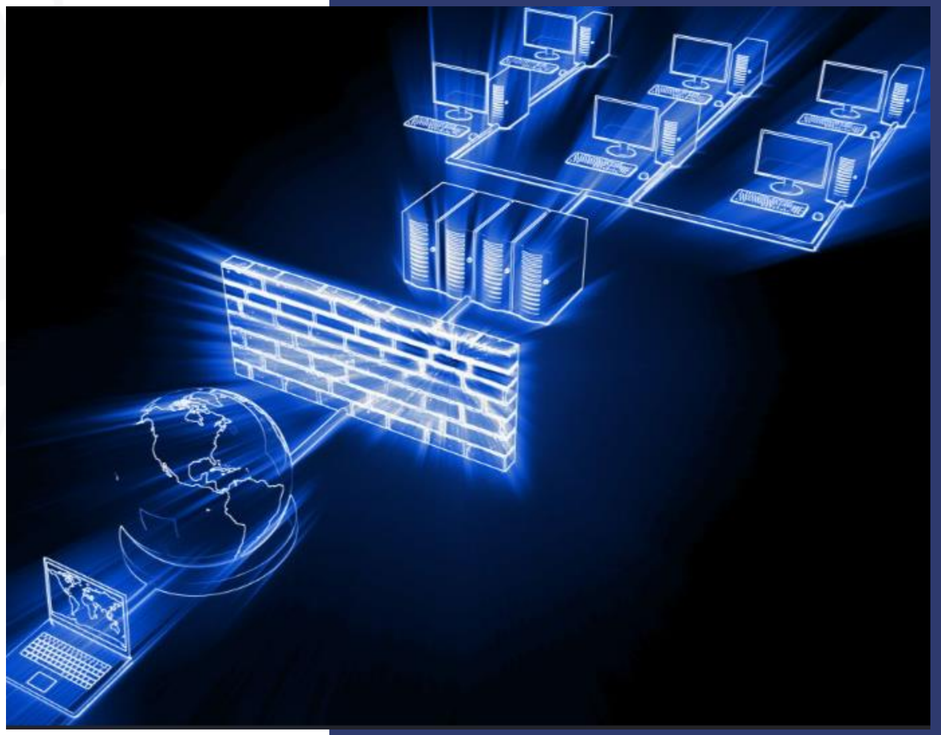
Types of Computer Network Architecture

Client-Server Architecture:

- A type of computer network architecture in which nodes can be servers or clients. The server node can manage the client node behaviors.

Peer-to-Peer Architecture:

- P2P (Peer-to-Peer) Architecture, is not based on the concept of a central server. Each device is free for working as either client or server.



Terminologies of Computer Networks

Nodes:
- Nodes are devices that are connected to a network. These can include computers, servers, printers, routers, switches, and other devices.

Protocol:
- A protocol is a set of rules and standards that govern how data is transmitted over a network. Examples of protocols include TCP/IP, HTTP, and FTP.

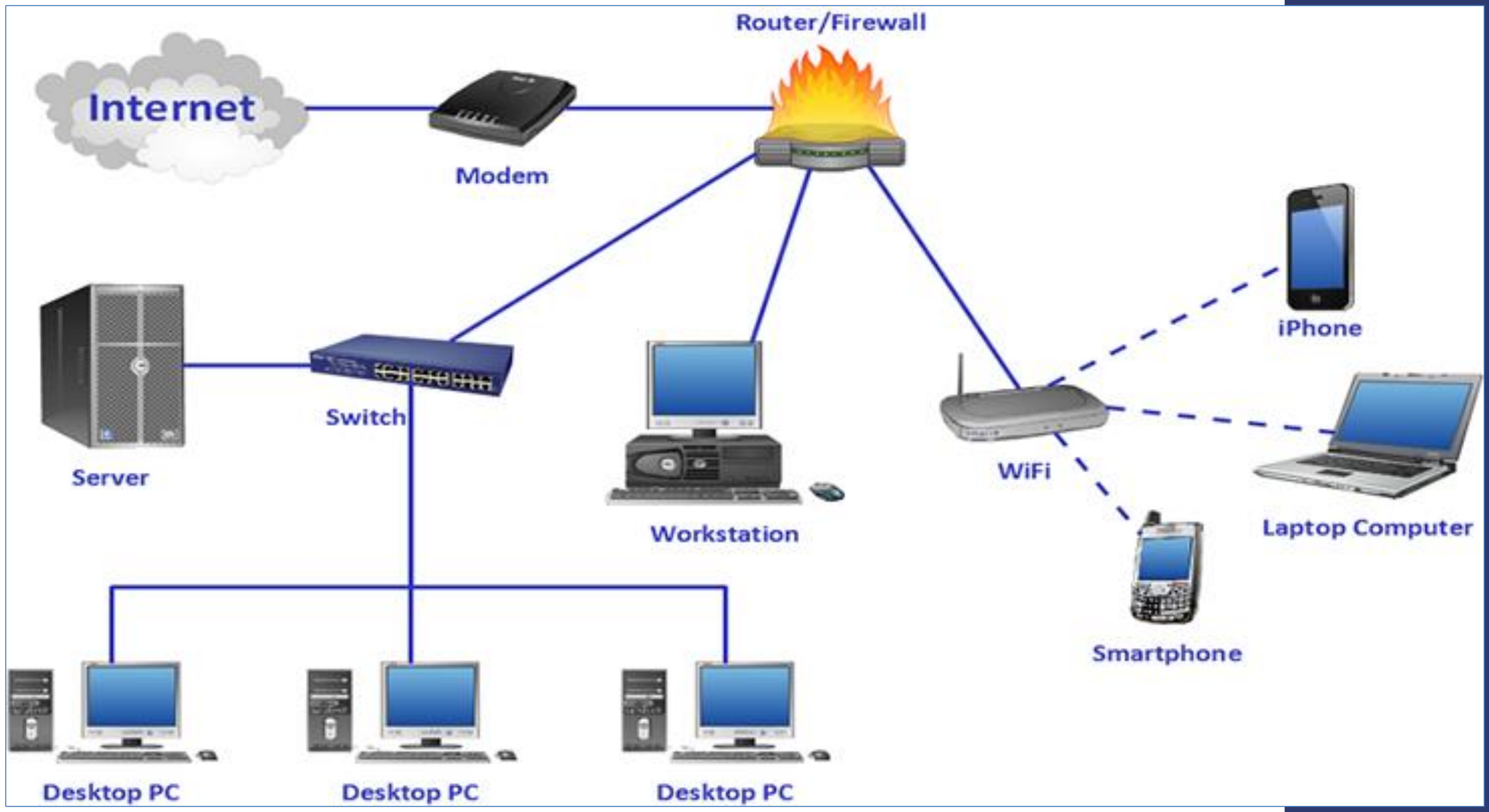
Topology:
- Network topology refers to the physical and logical arrangement of nodes on a network. The common network topologies include bus, star, ring, mesh, and tree.

IP Address:
- An IP address is a unique numerical identifier that is assigned to every device on a network. IP addresses are used to identify devices and enable communication between them.

DNS:
- The Domain Name System (DNS) is a protocol that is used to translate human-readable domain names (such as www.google.com) into IP addresses that computers can understand.

Firewall:
- A firewall is a security device that is used to monitor and control incoming and outgoing network traffic. Firewalls are used to protect networks from unauthorized access and other security threats.

Computer Network Devices



NIC and MAC

A NIC (Network Interface Card) is the physical hardware component that allows a device to connect to a network, while a MAC (Media Access Control) address is the unique, permanent identifier embedded in that NIC.

The Media Access Control (MAC) address or physical address is encoded on the network card chip which is assigned by the IEEE to identify a network card uniquely. The MAC address is stored in the PROM (Programmable read-only memory).

There are two types of NIC:



- **Wired NIC:** The Wired NIC is present inside the motherboard. Cables and connectors are used with wired NIC to transfer data.
- **Wireless NIC:** The wireless NIC contains the antenna to obtain the connection over the wireless network. For example, laptop computer contains the wireless NIC

Hub

- A Hub is a hardware device that divides the network connection among multiple devices. When computer requests for some information from a network, it first sends the request to the Hub through cable.

- Hub will broadcast this request to the entire network. All the devices will check whether the request belongs to them or not. If not, the request will be dropped.

- The process used by the Hub consumes more bandwidth and limits the amount of communication.

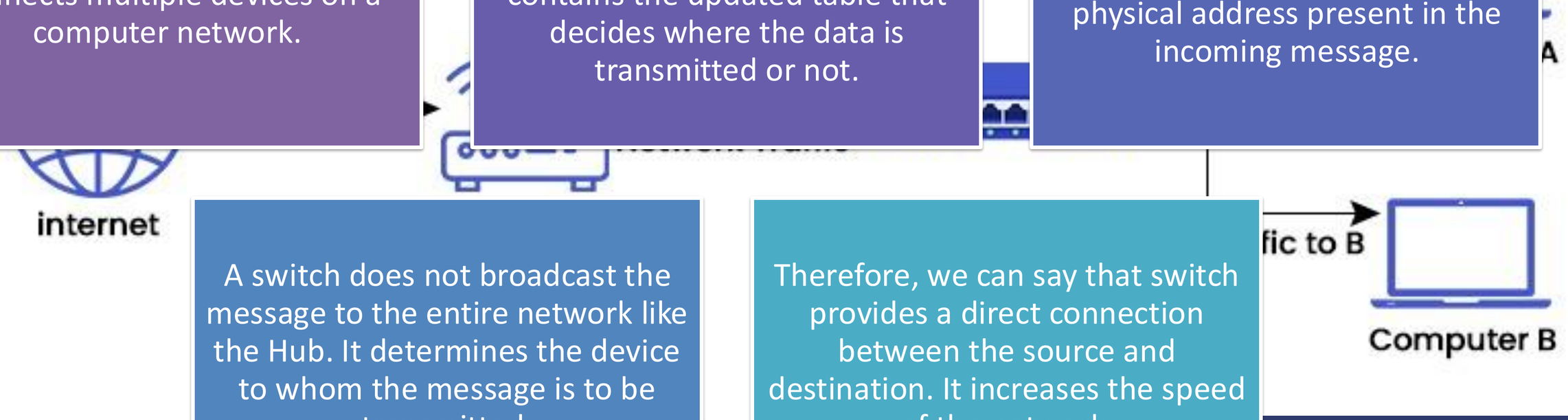
- Nowadays, the use of hub is obsolete, and it is replaced by more advanced computer network components such as Switches, Routers.

Switch

A switch is a hardware device that connects multiple devices on a computer network.

A switch contains more advanced features than Hub. The Switch contains the updated table that decides where the data is transmitted or not.

Switch delivers the message to the correct destination based on the physical address present in the incoming message.



A switch does not broadcast the message to the entire network like the Hub. It determines the device to whom the message is to be transmitted.

Therefore, we can say that switch provides a direct connection between the source and destination. It increases the speed of the network.

Router

A router is a hardware device which is used to connect a LAN with an internet connection. It is used to receive, analyze and forward the incoming packets to another network. A router works in a **Layer 3 (Network layer)** of the Open System Interconnection (OSI) Reference model. A router forwards the packet based on the information available in the routing table. It determines the best path from the available paths for the transmission of the packet.



Advantages Of Router:

- **Security:** The information which is transmitted to the network will traverse the entire cable, but the only specified device which has been addressed can read the data.
- **Reliability:** If the server has stopped functioning, the network goes down, but no other networks are affected that are served by the router.
- **Performance:** Router enhances the overall performance of the network. Suppose there are 24 workstations in a network generates a same amount of traffic. This increases the traffic load on the network. Router splits the single network into two networks of 12 workstations each, reduces the traffic load by half.

Types of Computer Networks

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wireless Local Area Network (WLAN)
- Campus Area Network (CAN)
- Wide Area Network (WAN)
- Storage Area Network (SAN)
- Enterprise Private Area Network
- Virtual Private Network (VPN)

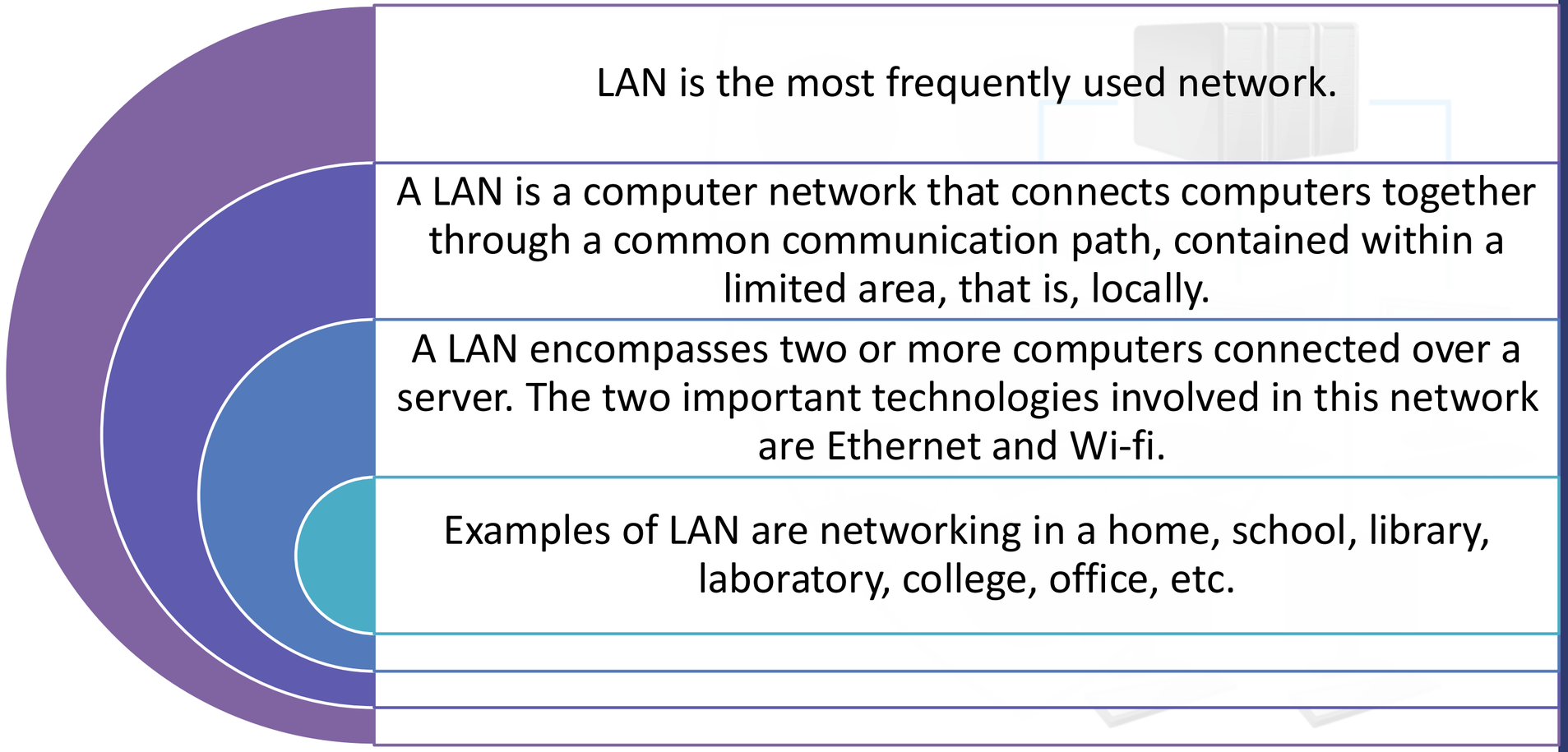


Personal Area Network [PAN]

- PAN is the most basic type of computer network.
- This network is restrained to a single person, that is, communication between the computer devices is centred only to an individual's work space.
- PAN offers a network range of 10 meters from a person to the device providing communication.
- Examples of PAN are USB, computer, phone, tablet, printer, PDA, etc.



Local Area Network [LAN]



LAN is the most frequently used network.

A LAN is a computer network that connects computers together through a common communication path, contained within a limited area, that is, locally.

A LAN encompasses two or more computers connected over a server. The two important technologies involved in this network are Ethernet and Wi-fi.

Examples of LAN are networking in a home, school, library, laboratory, college, office, etc.

Wide Area Network [WAN]

- WAN is a type of computer network that connects computers over a large geographical distance through a shared common path.
- It is not restrained to a single location but extends over many locations.
- WAN can also be defined as a group of local area networks that communicate with each other.
- The most common example of WAN is the Internet.



Wireless Local Area Network [WLAN]

WLAN is a type of computer network that acts as a local area network but makes use of wireless network technology.

This network doesn’t allow devices communicating over physical cables like in LAN but allows devices to communicate wirelessly.

The most common example of WLAN is Wi-Fi.



Metropolitan Area Network [MAN]

A MAN is larger than a LAN but smaller than a WAN.

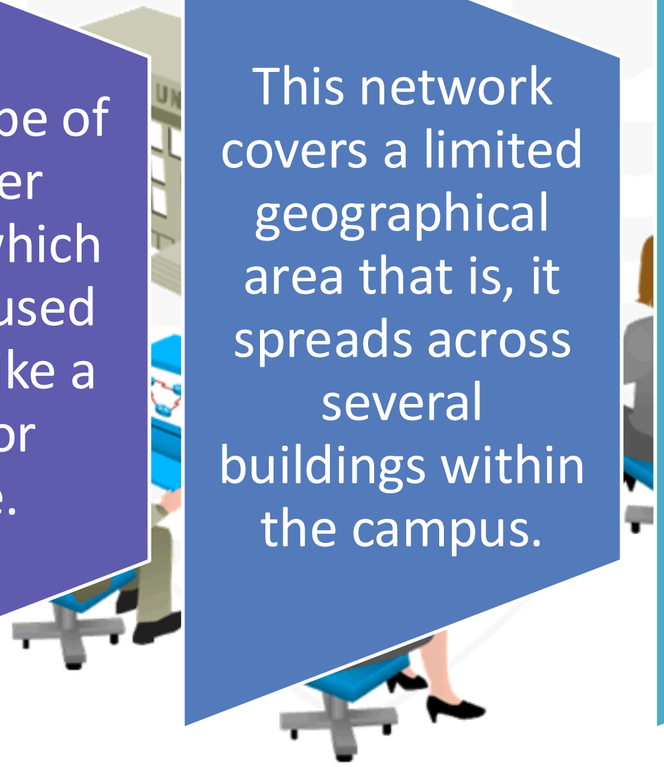
This is the type of computer network that connects computers over a geographical distance through a shared communication path over a city, town or metropolitan area.

Examples of MAN are networking in towns, cities, a single large city, large area within multiple buildings, etc.

Campus Area Network [CAN]

CAN is bigger than a LAN but smaller than a MAN.

This is a type of computer network which is usually used in places like a school or college.



This network covers a limited geographical area that is, it spreads across several buildings within the campus.

Examples of CAN are networks that cover schools, colleges, buildings, etc.

Storage Area Network [SAN]

SAN is a type of computer network that is high speed and connects groups of storage devices to several servers.

This network does not depend on LAN or WAN. Instead, a SAN moves the storage resources from the network to its own high-powered network.

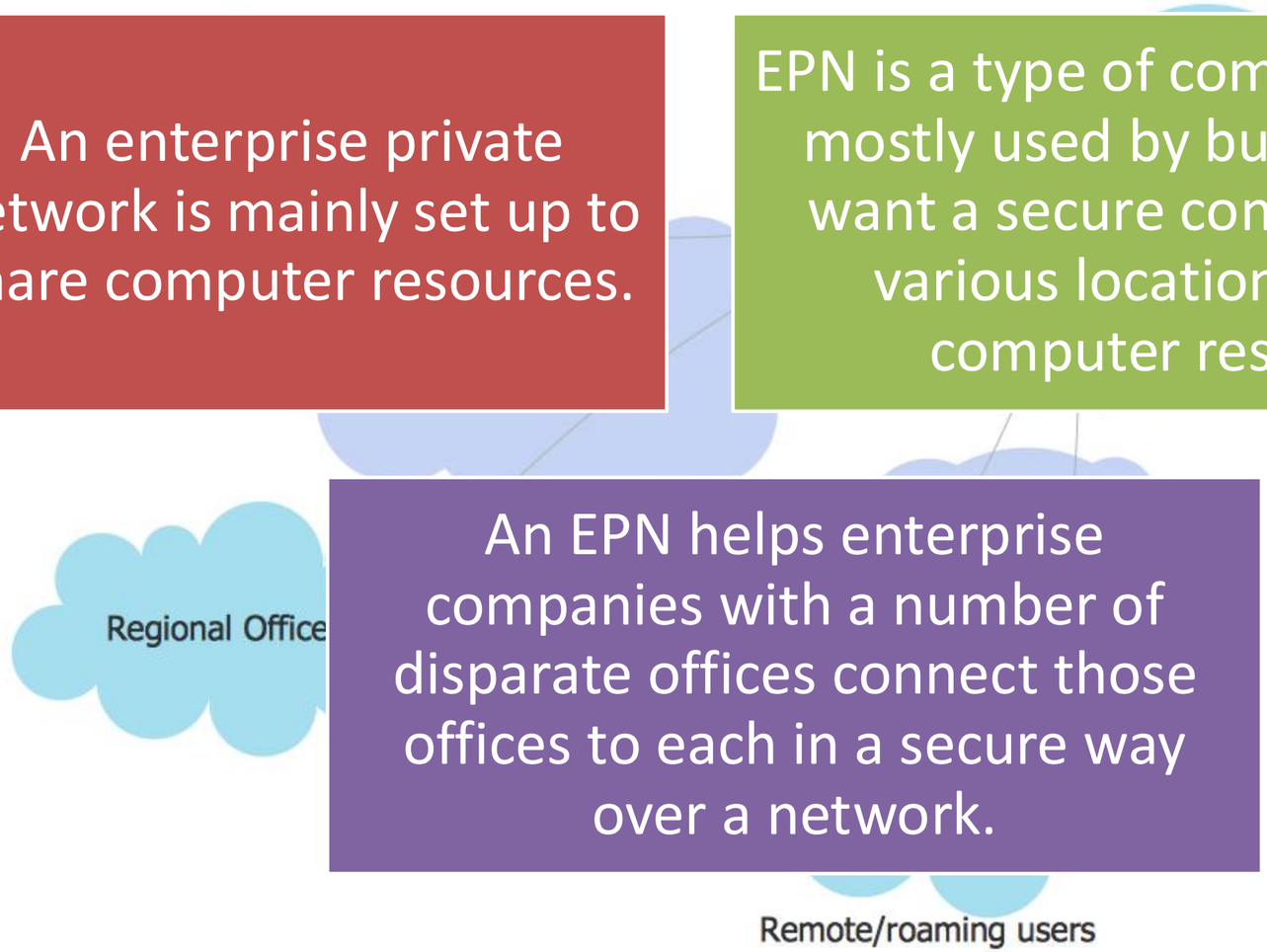
A SAN provides access to block-level data storage.

Examples of SAN are a network of disks accessed by a network of servers.

Enterprise Private Network [EPN]

An enterprise private network is mainly set up to share computer resources.

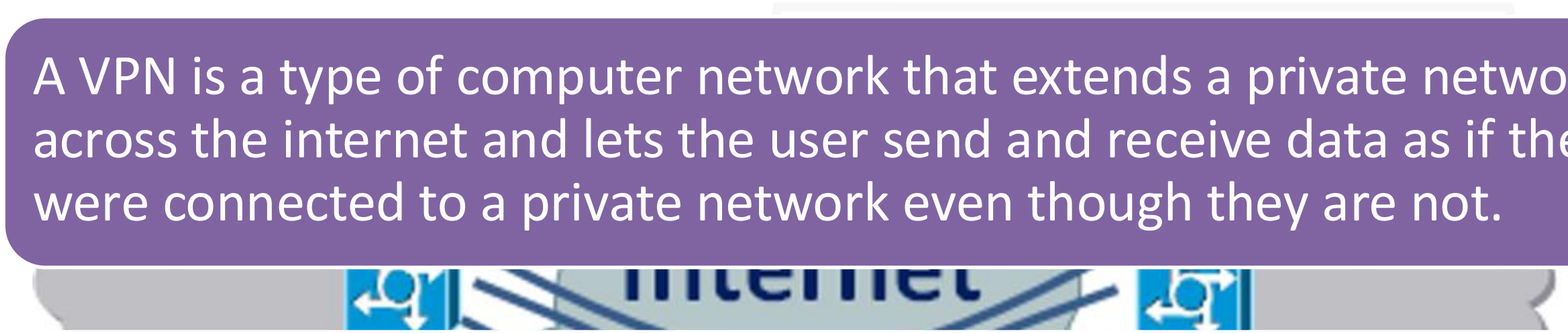
EPN is a type of computer network mostly used by businesses that want a secure connection over various locations to share computer resources.



An EPN helps enterprise companies with a number of disparate offices connect those offices to each in a secure way over a network.

Virtual Private Network [VPN]

A VPN is a type of computer network that extends a private network across the internet and lets the user send and receive data as if they were connected to a private network even though they are not.



Through a virtual point to point connection users can access a private network remotely.

VPN protects you from malicious sources by operating as a medium that gives you protected network connection.

Network Protocols

Transmission Control Protocol/Internet Protocol (TCP/IP):

- TCP/IP is the foundational protocol suite of the internet, enabling reliable communication. It ensures data is delivered reliably and in order and IP routes data packets to their destination based on IP addresses.

Hypertext Transfer Protocol (HTTP) and HTTPS:

- HTTP and HTTPS protocols used for transmitting web pages. **In HTTP** communication is unsecured and in **HTTPS** secured communication using SSL/TLS encryption.

Simple Mail Transfer Protocol (SMTP):

- SMTP protocol is used to send email. SMTP protocol works with other protocols like POP3 and IMAP for email retrieval.

File Transfer Protocol (FTP):

- FTP protocol used for transferring files between computers. Includes commands for uploading, downloading, and managing files on a remote server.

Intranet vs Extranet

Intranet

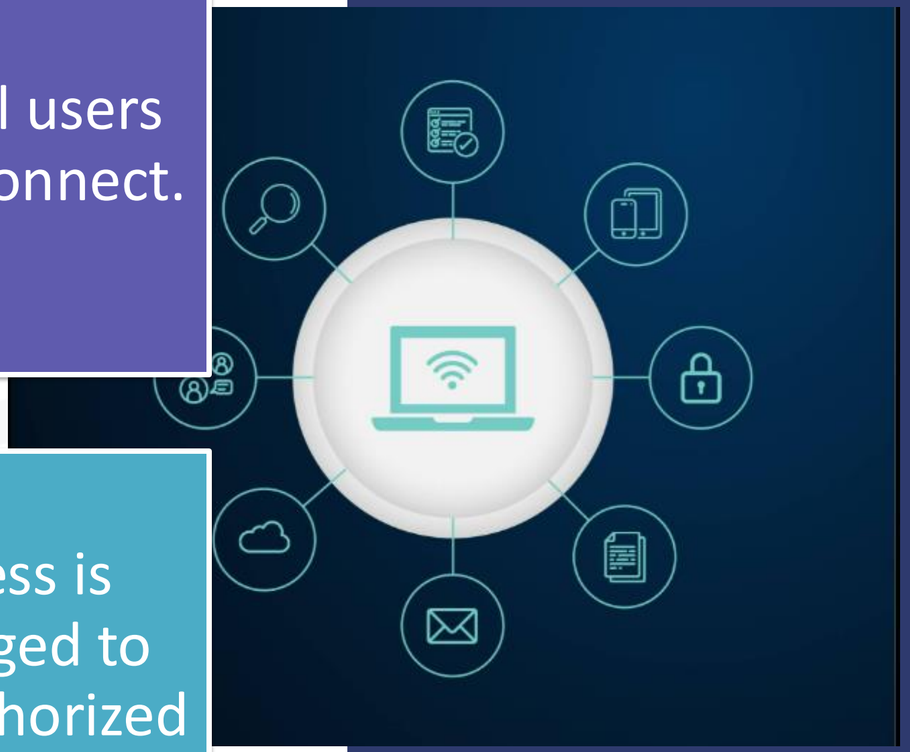
Extranet



Intranet

Intranet is a computer network which is controlled and administered by a single entity such as company or organization.

In Intranet, external users are not allowed to connect.



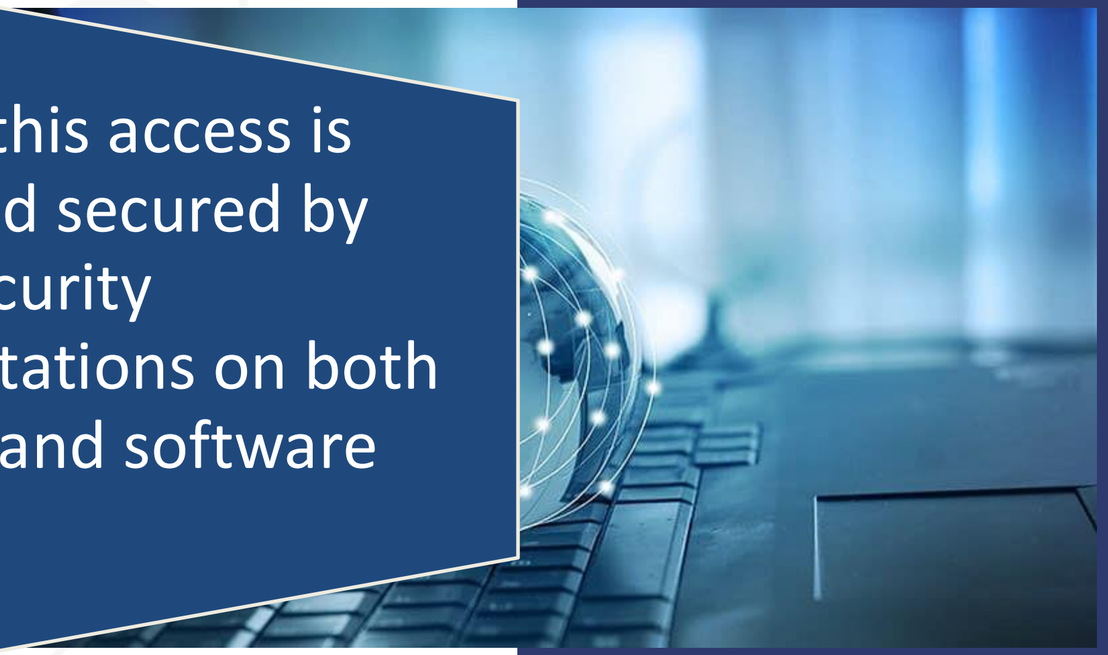
Usually in this network, proper authorization is required to access any resources.

Further each access is monitored and logged to ensure that only authorized users get access.

Extranet

An Extranet is an extended Intranet where external users are allowed to access a small portion of network.

Normally this access is filtered and secured by several security implementations on both hardware and software levels.



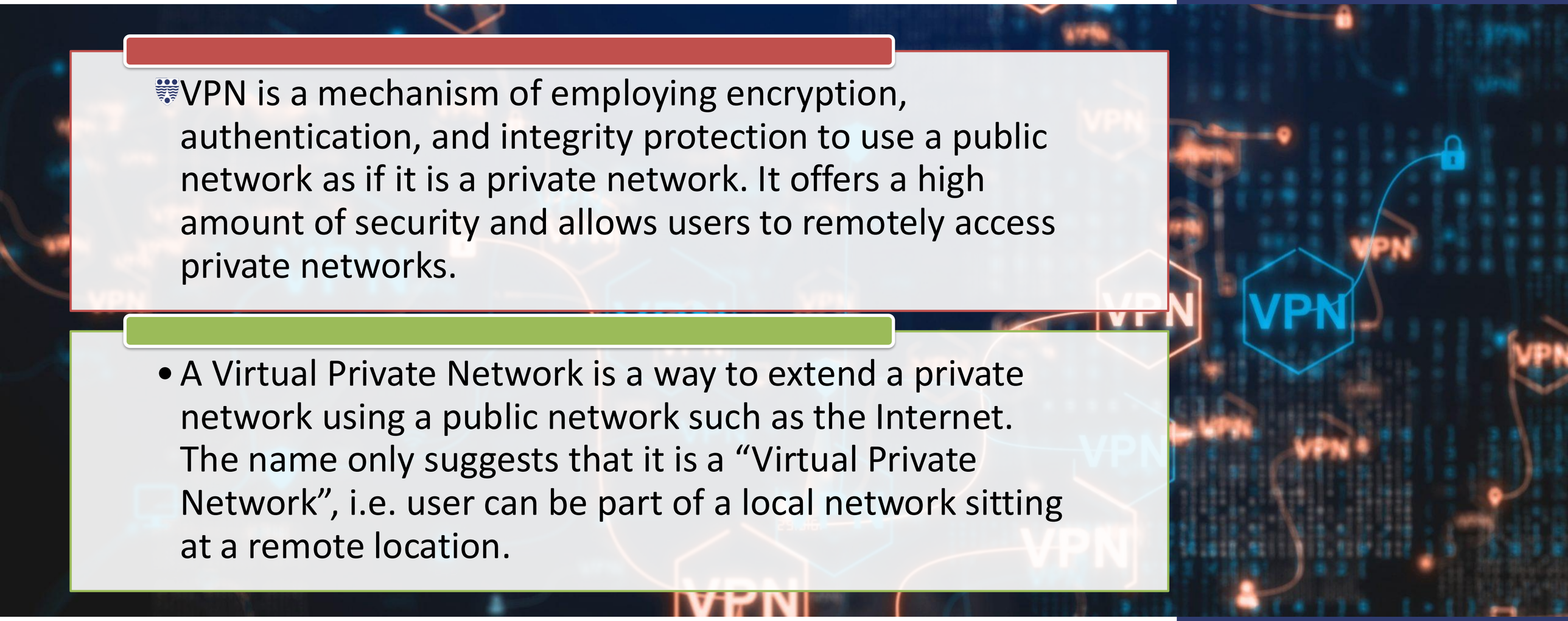
VPN vs Client/Server

VPN

Client/Server



Virtual Private Network [VPN]



VPN is a mechanism of employing encryption, authentication, and integrity protection to use a public network as if it is a private network. It offers a high amount of security and allows users to remotely access private networks.

- A Virtual Private Network is a way to extend a private network using a public network such as the Internet. The name only suggests that it is a “Virtual Private Network”, i.e. user can be part of a local network sitting at a remote location.

Client/Server Network

In a client/server network a dedicated computer known as server provides sharing resources. All other computers know as clients are used to access the shared resources.

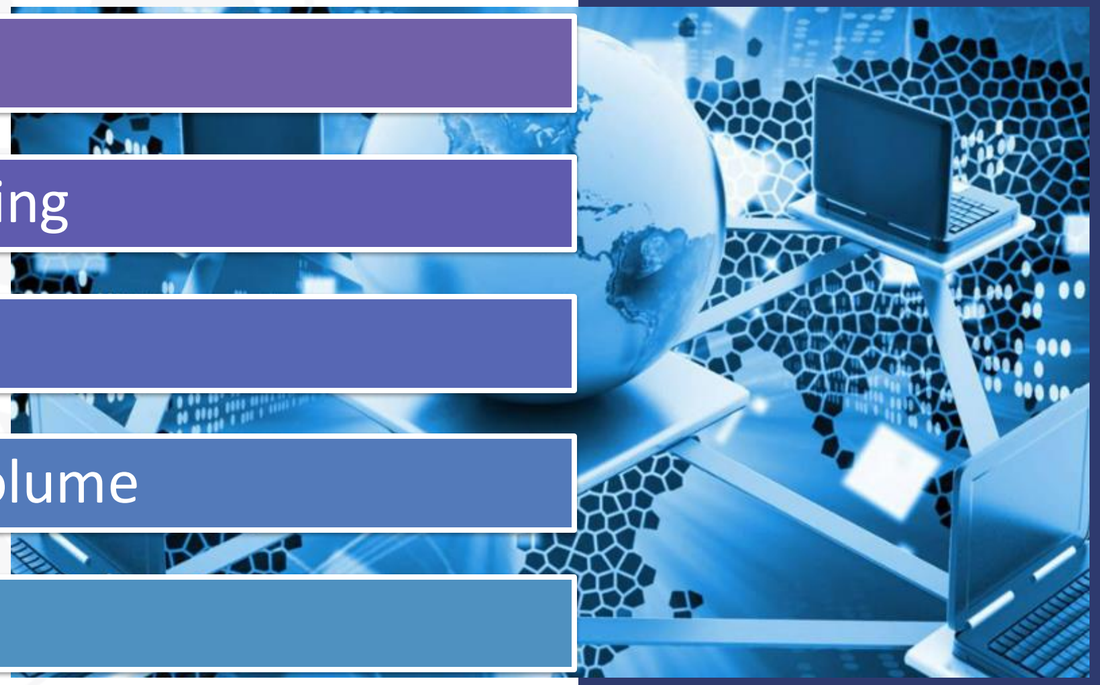


This type of network is commonly used in company environment. It provides great security features but requires special hardware and software to setup.

In the client-server architecture, when the client computer sends a request for data to the server through the internet, the server accepts the requested process and delivers the data packets requested back to the client

Benefits of Computer Networks

- Easy Sharing of Data & Hardware Resources
- Easy Sharing of Software
- Easy to Decentralize Data Processing
- Easy to Communicate
- Improve Storage Efficiency and Volume
- Cut Cost on Software
- Secure Valuable Information



Computer Network Security

Firewalls:	Devices or software that monitor and control incoming and outgoing network traffic based on security rules.
Encryption:	The process of encoding data to prevent unauthorized access. Commonly used in VPNs, HTTPS, and secure email.
Intrusion Detection Systems (IDS):	Tools that monitor network traffic for suspicious activity and potential threats.
Access Control:	Mechanisms that restrict access to network resources based on user identity and role.
Regular Updates and Patching:	Keeping software and hardware up to date to protect against vulnerabilities.

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Thank You

Q & A